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Boundary-Integral Type Neural Network (BINN) for Flow Problems in Anisotropic Reservoirs

Xiang Rao, School of Petroleum Engineering, Yangtze University, Wuhan, Hubei, China / State Key Laboratory of Low Carbon Catalysis and Carbon Dioxide Utilization, Yangtze University, Wuhan, China / Western Research Institute, Yangtze University, Karamay, China / Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia; Yina Liu and Qirun Fu, Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia; Xupeng He and Hyung Kwak, Saudi Aramco, Dhahran, Saudi Arabia; Hui Zhao, School of Petroleum Engineering, Yangtze University, Wuhan, Hubei, China / State Key Laboratory of Low Carbon Catalysis and Carbon Dioxide Utilization, Yangtze University, Wuhan, China / Western Research Institute, Yangtze University, Karamay, China; Hussein Hoteit, Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

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Abstract

Accurate simulation of fluid flow in anisotropic reservoirs is challenging for traditional numerical methods and physics-informed neural networks (PINNs) due to issues with complex geometries, singularities, and scalability. This study introduces a Boundary-Integral Neural Network (BINN) framework that merges boundary integral equations (BIEs) with deep learning for 2D single-phase flow in anisotropic reservoirs. By converting governing partial differential equations into boundary-only integral equations via Green's functions, BINN focuses neural network learning on boundary unknowns (e.g., pressure and normal derivatives), reconstructing domain solutions through boundary integrals. This reduces dimensionality, avoids global sampling costs of PINNs, enforces boundary conditions analytically, and handles singularities near wells/fractures via Green's function kernels. Three examples validate BINN: flow with full-tensor permeability in a rectangle, single-well-induced pressure singularities, and multi-well flow in an irregular trapezoidal domain. Results align closely with boundary element method (BEM) solutions, with RMSEs of 5.17×10^{-4} MPa for single-well cases and $< 10^{-3}$ MPa for complex scenarios. BINN shows stable convergence across geometric irregularities and anisotropic effects, demonstrating robustness in capturing flow physics under coupled influences of tensor permeability, source-sink terms, and complex boundaries. This work positions BINN as a promising meshless approach for anisotropic reservoir simulation.

Keywords: Flow in porous media, Boundary integral equations (BIEs), Physics-informed neural networks (PINNs), Boundary-Integral Neural Networks (BINNs)

Introduction

In the realm of hydrocarbon exploration and production, there is an escalating need for highly accurate and efficient reservoir flow simulations. Traditional numerical methods, which rely on spatial discretization meshes (Rao et al., 2024a; Rao et al., 2025a; Liu et al., 2025), encounter significant challenges when dealing with anisotropic reservoirs where permeability is characterized by a tensor. These methods struggle to effectively handle complex geometries and highly nonlinear scenarios, leading to substantial computational burdens, intricate mesh generation issues, and numerical dispersion problems (Rao et al., 2025b). Although meshless methods can avoid the complexities associated with mesh generation, they still grapple with high computational costs and numerical dispersion issues (Rao et al., 2024a; Rao et al., 2022; Liu et al., 2023).

In recent years, deep learning frameworks, especially Physics-Informed Neural Networks (PINNs), have emerged as innovative approaches. PINNs embed physical laws as residuals in loss functions, enabling meshless solutions to partial differential equations (PDEs) (Raissi et al., 2019). Leveraging the function approximation capabilities of neural networks, they directly learn solution mappings from data and governing equations, potentially eliminating the need for spatial discretization meshes that traditional methods rely on (Harp et al., 2021; Fuks and Tchelep, 2020; Liu et al., 2025;). However, when applied to reservoir-scale flow problems in anisotropic media, PINNs encounter significant drawbacks. Their global sampling methods cause training costs to skyrocket as the computational domain increases, and they often show reduced accuracy near boundary singularities, such as wellbore or fracture regions (Chen et al., 2022; Fan et al., 2022; Rao, 2023).

To address these challenges, Boundary Integral Neural Networks (BINNs) have been developed (Sun et al. 2023; Lin et al., 2023). BINNs combine the mathematical principles of Boundary Integral Equations (BIE) with deep learning architectures, converting PDEs into boundary-only integral equations using Green's functions (Rao et al., 2018; Rao et al., 2019). In this framework, neural networks only need to learn boundary unknowns, such as potential functions or their normal derivatives, and the domain solution is reconstructed through boundary integrals. Compared with PINNs' global sampling-based solution approach, BINNs have significant advantages in efficiency and robustness. Firstly, by solving BIEs, BINNs reduce dimensionality, requiring only boundary sampling, which greatly improves scalability for large-scale simulations of anisotropic reservoirs. Secondly, they enforce boundary conditions analytically, overcoming the approximation errors of PINN penalty methods and ensuring the mathematical consistency of the solution. Thirdly, the Green's function kernel in boundary integral equations naturally adapts to the characteristics of fundamental solutions, allowing for accurate capture of solution behavior in singular regions, like well locations or fracture tips, without the need for special treatment.

In recent years, boundary-integral neural networks (BINNs) have witnessed substantial advancements in both theoretical frameworks and practical implementations. Sun et al. (2023) pioneered BINNs for solving boundary value problems in computational mechanics, addressing complex geometries, infinite/semi-infinite domains, and heterogeneous material systems. Lin et al. (2023) developed BINet, a hybrid approach fusing boundary integral equations with neural networks, enabling the solution of partial differential equations (PDEs) in both bounded and unbounded spaces. Zhang et al. (2024) made the first foray into applying BINNs to two-dimensional elastostatic and piezoelectric problems, while Schmid et al. (2025) leveraged these networks for acoustic radiation prediction. By integrating boundary integral equation residuals into the loss function, their model achieved data-driven predictions of acoustic fields from noisy boundary data, outperforming conventional boundary element methods for both interior and exterior acoustic problems. Qu et al. (2024) introduced a novel BINN framework for acoustic radiation and scattering analysis, demonstrating comparable precision to physics-informed neural networks (PINNs) with fewer collocation points and network parameters, attributed to the semi-analytic nature of boundary integral equations (BIEs). Zhang et al. (2025) extended BINNs to linear elastic fracture mechanics, incorporating fracture mechanics asymptotics to enhance accuracy near crack tips, where stress singularities and steep

gradients occur. Damiano and Waterschoot (2025) further advanced the field by developing a boundary integral network for sound field reconstruction, using the Kirchhoff-Helmholtz BIE to model spatial sound fields and a shallow neural network to retrieve boundary pressure distributions for interior acoustic pressure prediction. These collective advancements highlight BINNs as a powerful paradigm for integrating physical principles with data-driven learning in complex engineering problems.

Despite these achievements, the application of BINNs to subsurface flow modeling in anisotropic reservoirs is still largely unexplored. This study aims to investigate the basic application of BINNs in two-dimensional single-phase flow problems in anisotropic reservoirs, constructing a boundary integral-based neural network model for flow simulation. The proposed method seeks to provide a new methodological perspective for numerical modeling of hydrocarbon reservoirs with anisotropic properties.

Methodology

The governing equation

The governing equations for subsurface porous media seepage mainly include the fluid mass conservation equation and Darcy's law. For steady-state flow conditions with source or sink terms considered, the fluid mass conservation equation can be expressed as:

$$\nabla \cdot \mathbf{v} = Q(\mathbf{x}), \mathbf{x} \in \Omega \subset \mathbb{R}^2 \quad (1)$$

where $\mathbf{v} = (v_x, v_y)_T$ is the 2D seepage velocity; $Q(\mathbf{x})$ denotes the source/sink strength per unit volume.

When the permeability is a tensor $\mathbf{K}$, Darcy's law is described as:

$$\mathbf{v} = -\frac{1}{\mu} \mathbf{K} \nabla p \quad (2)$$

where p is the fluid pressure; μ is the fluid viscosity; $\mathbf{K} = \begin{pmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{pmatrix}$, in which, $K_{xy} = K_{yx}$.

$$\nabla \cdot (\mathbf{K} \nabla p) = Q(\mathbf{x}), \mathbf{x} \in \Omega \subset \mathbb{R}^2 \quad (3)$$

In reservoir fluid flow, the source-sink term $Q(\mathbf{x})$ in Eq. (3) is typically composed of singular source-sink terms induced by injection-production wells, which can be expressed as:

$$Q(\mathbf{x}) = \sum_{\eta=1}^{n_s} Q_{\eta} \delta(\mathbf{x} - \mathbf{x}_{s\eta}) \quad (4)$$

where n_s is the number of the source-sink terms, Q_{η} and $\mathbf{x}_{s\eta}$ are the strength and location coordinates of the η -th source-sink term, respectively.

Expanded in the Cartesian coordinate system as:

$$\frac{\partial}{\partial x} \left(K_{xx} \frac{\partial p}{\partial x} + K_{xy} \frac{\partial p}{\partial y} \right) + \frac{\partial}{\partial y} \left(K_{yx} \frac{\partial p}{\partial x} + K_{yy} \frac{\partial p}{\partial y} \right) = \sum_{\eta=1}^{n_s} Q_{\eta} \delta(\mathbf{x} - \mathbf{x}_{s\eta}) \quad (5)$$

This flow problem generally has Dirichlet boundary conditions representing constant pressure boundaries or Neumann boundary conditions representing sealed or constant flow rates.

$$\text{Dirichlet boundary condition: } p(\mathbf{x}) = f(\mathbf{x}), \mathbf{x} \in \Gamma_{DB},$$

$$\text{Neumann boundary condition: } \frac{\partial p(\mathbf{x})}{\partial n_{\mathbf{K}}} = (\mathbf{K} \nabla p) \cdot \mathbf{n} = g(\mathbf{x}), \mathbf{x} \in \Gamma_{NB}, \quad (6)$$

where $f(x)$ and $g(x)$ represents the known functions, and Γ_{db} denotes the Dirichlet boundary, Γ_{nb} denotes the Neumann boundary.

$\frac{\partial}{\partial n_{\mathbf{K}}}$ denotes the normal derivative operator based on the tensor $\mathbf{K}$, when $\frac{\partial}{\partial n_{\mathbf{K}}}$ act on an arbitrary function $h(\mathbf{x})$, there is:

$$\frac{\partial}{\partial n_{\mathbf{K}}} h(\mathbf{x}) = \mathbf{n} \cdot (\mathbf{K} \nabla h) \quad (7)$$

Boundary integral equations (BIEs)

For $\nabla \cdot (\mathbf{K} \nabla p) = 0$, it is necessary to construct the Green's function corresponding to the tensor permeability. Assuming that $\mathbf{K}$ is a constant tensor, the equation can be transformed into an isotropic form through coordinate transformation (such as principal axis transformation), but here we directly derive the general form.

Define the fundamental solution under tensor permeability as $G(\mathbf{x}, \mathbf{a})$, satisfying:

$$\nabla \cdot (\mathbf{K} \nabla G(\mathbf{x}, \mathbf{a})) = -\delta(\mathbf{x} - \mathbf{a}) \quad (8)$$

In two-dimensional cases, when $\mathbf{K}$ is a constant tensor, $G(\mathbf{x}, \mathbf{a})$ is derived as:

$$G(\mathbf{x}, \mathbf{a}) = \frac{1}{2\pi\sqrt{\det(\mathbf{K})}} \ln(r_{\mathbf{K}}) \quad (9)$$

where $\det(\mathbf{K}) = K_{xx}K_{yy} - K_{xy}^2$, $r_{\mathbf{K}} = \sqrt{r^T \mathbf{K}^{-1} r}$, $r = \mathbf{x} - \mathbf{a}$.

Applying Green's second formula, for any interior point $\mathbf{a}$, we have:

$$p(\mathbf{a}) = \int_{\Gamma} \left(G(\mathbf{x}, \mathbf{a}) \frac{\partial p}{\partial n_{\mathbf{K}}}(\mathbf{x}) - p(\mathbf{x}) \frac{\partial G}{\partial n_{\mathbf{K}}}(\mathbf{x}, \mathbf{a}) \right) d\Gamma \quad (10)$$

where, $\frac{\partial}{\partial n_{\mathbf{K}}} = \mathbf{n} \cdot (\mathbf{K} \nabla G)$.

When the observation point ξ is on the boundary Γ , the geometric coefficient $c(\xi)$ is introduced to obtain the boundary integral equation of Eq. (3):

$$c(\xi)p(\xi) + \int_{\Gamma} p(\mathbf{x}) \frac{\partial G}{\partial n_{\mathbf{K}}}(\mathbf{x}, \xi) d\Gamma = \int_{\Gamma} G(\mathbf{x}, \xi) \frac{\partial p}{\partial n_{\mathbf{K}}}(\mathbf{x}, \xi) d\Gamma + \sum_{\eta=1}^{n_s} Q_{\eta} G(\mathbf{x}_{s\eta}, \xi) \quad (11)$$

where, if ξ is the interior point of Ω , $c(\xi) = 1$, if ξ is a point on the smooth boundary, $c(\xi) = \frac{1}{2}$, and if ξ is a corner point on the boundary, $c(\xi) = \theta$, in which θ is the included angle at the corner point.

Discretization of BIEs

We implement the discretization of the boundary integral equation into a linear system via constant elements. As illustrated in Fig. 1, the boundary Γ is partitioned into n elements $\Gamma_j (j=1, 2, \dots, n)$. For constant elements, both the function values and normal derivative values are assumed to remain uniform across each boundary element, taking the same values as those at the corresponding nodal points.

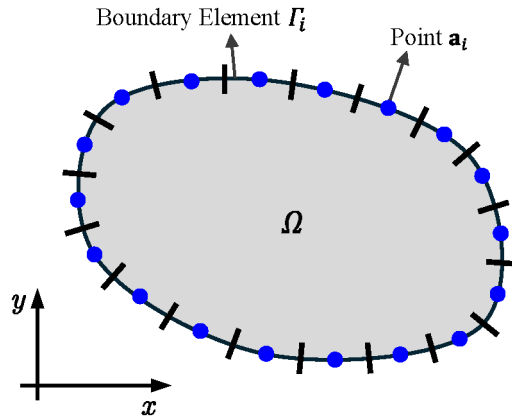


Figure 1—Discretization of computational domain boundary

Let $\xi = \mathbf{a}_i$, Eq. (10) can be discretized as:

$$c(\mathbf{a}_i)p(\mathbf{a}_i) + \sum_{j=1}^{n_e} \int_{\Gamma_j} p(\mathbf{x}) \frac{\partial G}{\partial n_{\mathbf{K}}}(\mathbf{x}, \mathbf{a}_i) d\Gamma - \sum_{j=1}^{n_e} \int_{\Gamma_j} G(\mathbf{x}, \mathbf{a}_i) \frac{\partial p}{\partial n_{\mathbf{K}}}(\mathbf{x}, \mathbf{a}_i) d\Gamma - \sum_{\eta=1}^{n_s} Q_{\eta} G(\mathbf{x}_{s\eta}, \mathbf{a}_i) = 0 \quad (12)$$

where, $\mathbf{a}_i$ denotes the midpoint of boundary elements Γ_i .

Clarify the terms in Eq. (12). In this equation, when $j \neq i$, the integrals over the boundary elements are regular integrals, which can be evaluated using a high-order Gaussian quadrature formula. Conversely, when $j = i$, the integrals become singular integrals, which require specialized techniques such as analytical methods or adaptive integration for computation

$$\begin{aligned} \mathbf{H}_{ij} &= \int_{\Gamma_j} p(\mathbf{x}) \frac{\partial G}{\partial n_{\mathbf{K}}}(\mathbf{x}, \mathbf{a}_i) d\Gamma + c(\mathbf{a}_i) \delta_{ij} \\ \mathbf{M}_{ij} &= \int_{\Gamma_j} G(\mathbf{x}, \mathbf{a}_i) d\Gamma \\ \mathbf{F}_i &= \sum_{\eta=1}^{n_s} Q_{\eta} G(\mathbf{x}_{s\eta}, \mathbf{a}_i) \end{aligned} \quad (13)$$

in which, δ_{ij} is the Kronecker function.

Using Eq. (12), Eq. (10) can be written as:

$$\sum_{j=1}^{n_e} \mathbf{H}_{ij} p_j - \sum_{j=1}^{n_e} \mathbf{M}_{ij} \left(\frac{\partial p}{\partial n_{\mathbf{K}}} \right)_j - \mathbf{F}_i = 0 \quad (14)$$

Write Eq. (13) in a matrix form:

$$\mathbf{H}\mathbf{p} - \mathbf{M} \frac{\partial \mathbf{p}}{\partial n_{\mathbf{K}}} - \mathbf{F} = 0 \quad (15)$$

where $\mathbf{p} = (p_1, p_2, \dots, p_{n_e})^T$, $\frac{\partial \mathbf{p}}{\partial n_{\mathbf{K}}} = \left(\left(\frac{\partial p}{\partial n_{\mathbf{K}}} \right)_1, \left(\frac{\partial p}{\partial n_{\mathbf{K}}} \right)_2, \dots, \left(\frac{\partial p}{\partial n_{\mathbf{K}}} \right)_{n_e} \right)^T$.

Neural networks and loss function of the BINNs

Similar to PINN, BINN employ a fully connected multi-layer perceptron (MLP) architecture, as shown in Fig. 2, to approximate the unknown boundary variables in BIEs. The network architecture comprises an input layer, several hidden layers, and an output layer. The input layer takes the coordinates $\mathbf{x} = (x, y)$, while the output layer yields the pressure p .

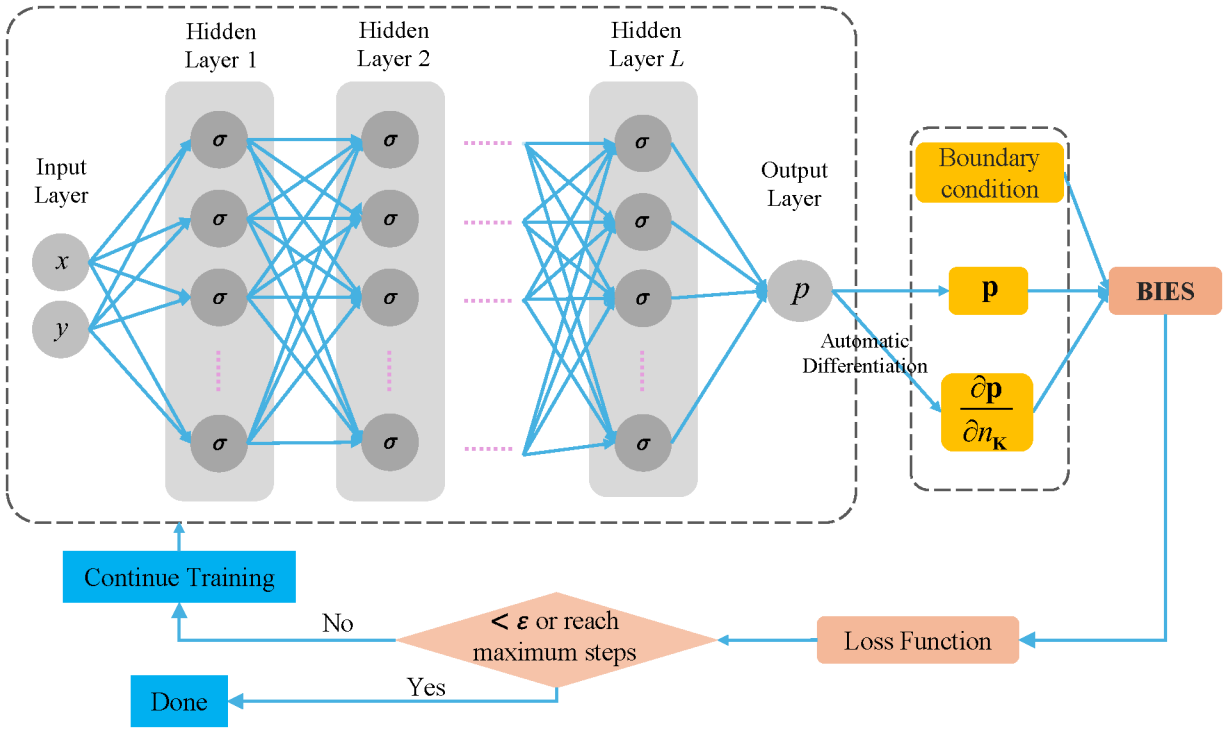


Figure 2—The framework of the boundary integrated neural network (BINN).

The output of the Z -th hidden layer of the network mapping shown in Fig. 2 is calculated by:

$$z_l = \sigma_l(\mathbf{W}_l z_{l-1} + \mathbf{b}_l), l = 1, 2, \dots, L \quad (16)$$

where $\mathbf{W}_l$ and $\mathbf{b}_l$ are the weight matrix and bias vector of the Z -th layer, respectively; σ_l denotes the activation function (e.g., ReLU, tanh). The output layer maps the hidden layer features to the boundary variables:

$$p(\mathbf{x}, \boldsymbol{\theta}) = \mathbf{W}_{L+1} z_L + \mathbf{b}_{L+1} \quad (17)$$

where $\boldsymbol{\theta} = \{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^{L+1}$, represents the network parameters.

By inputting the midpoint coordinates of each boundary element into the neural network shown in Fig. 2, we obtain a vector composed of pressure values at the centers of boundary elements $\mathbf{p}(\boldsymbol{\theta})$. Through the automatic differentiation of the neural network, we can derive

$\frac{\partial \mathbf{p}}{\partial n_{\mathbf{k}}}(\boldsymbol{\theta})$, which, combined with the boundary conditions in Eq. (6), enable the calculation of the mean squared error (MSE) in Eq. (15) during training as follows:

$$\mathcal{L}_{BIE}(\boldsymbol{\theta}) = \|\mathbf{H}\mathbf{p}(\boldsymbol{\theta}) - \mathbf{M} \frac{\partial \mathbf{p}}{\partial n_{\mathbf{k}}}(\boldsymbol{\theta}) - \mathbf{F}\|_2 \quad (18)$$

The training process of the neural network is terminated by checking whether $\mathcal{L}_{BIE}(\boldsymbol{\theta})$ is smaller than a threshold ε , or whether the number of training steps has reached the preset maximum.

Results and Discussions

To explore BINN application in reservoir seepage, two examples are designed in this section: Example 1 uses a non-diagonal permeability matrix to simulate seepage coupling effects for verifying BINN's anisotropic field characterization; Example 2 adds source-sink trms to study its accuracy in capturing complex seepage fields with a production well.

Example 1

This example focuses on exploring the capability of BINN in handling full-tensor permeability. When there is no source-sink term, the governing equation is expressed as:

$$\nabla \cdot (\mathbf{K} \nabla p) = 0 \quad (19)$$

where the permeability tensor $\mathbf{K}$ takes the form of a non-diagonal matrix:

$$\mathbf{K} = \begin{pmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{pmatrix} = \begin{pmatrix} 500 & 200 \\ 200 & 200 \end{pmatrix} mD \quad (20)$$

The computational domain is a rectangular region of 100 m×100 m, the boundary conditions are set as: constant pressure conditions of 10 MPa on the top and bottom boundaries, and impermeable boundaries on the left and right sides (Fig. 3), that is:

$$\begin{cases} p(x, 0) = 5MPa, & 0 \leq x \leq 100 \\ p(x, 100) = 10MPa, & 0 \leq x \leq 100 \\ v(0, y) = \frac{\partial p}{\partial n}(0, y) = 0, & 0 \leq y \leq 100 \\ v(100, y) = \frac{\partial p}{\partial n}(100, y) = 0, & 0 \leq y \leq 100 \end{cases} \quad (21)$$

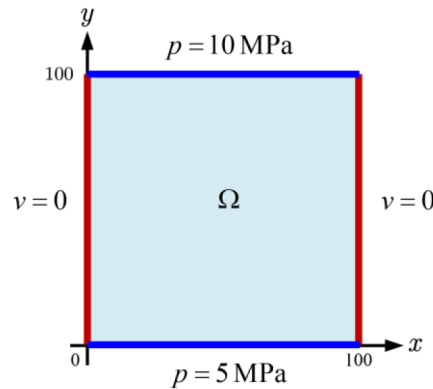


Figure 3—The reservoir domain, boundary elements, and interior points in Example 1.

The boundary is discretized using constant elements, with each side uniformly divided into 30 boundary elements, totaling 120 elements, and the midpoints of the elements are used as collocation points. To visualize the pressure distribution and verify calculation accuracy, 400 internal points are selected within the computational domain.

Table 1 shows the parameter configuration of Example 1. For pressure calculations at interior points, boundary integral formulas are employed to achieve the complete solution for the entire seepage field.

Table 1—Parameters of the neural networks and optimizers in Example 1

Model	Number of hidden layers	Number of neurons per hidden layer	Total number of parameters	Activation Function	Optimizer (Adams)
BINN	3	40	3441	Tanh	10000 iterations, learning rate 10^{-3}

The loss function evolution in Fig. 4 shows a rapid decline and subsequent stabilization during iterations, demonstrating that BINN exhibits excellent convergence in solving boundary integral equations. Its adaptive learning mechanism effectively approximates the boundary pressure distribution, benefiting from embedding the physical constraints of boundary integral equations into the neural network architecture and avoiding the complexity of singular integral handling in traditional numerical methods.

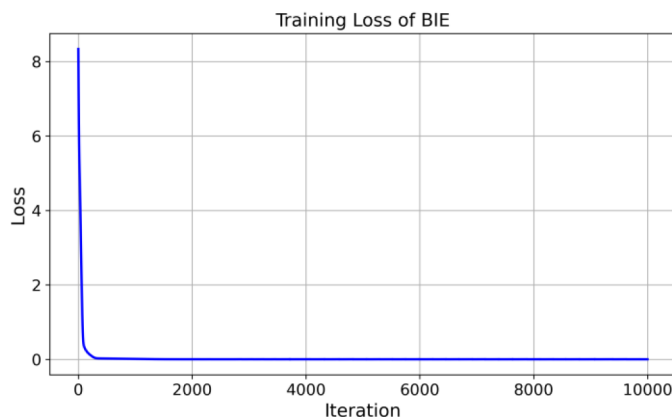


Figure 4—Loss function of BIE in Example 1.

The pressure distributions in Fig. 5 reveal that both BINN solutions and BEM reference solutions accurately characterize the pressure patterns under full-tensor permeability conditions. Owing to the non-diagonal components of the permeability tensor, the pressure contour lines exhibit a morphology distinct from isotropic scenarios, with high consistency in pressure distribution trends between the two methods. This consistency, particularly in accurately reflecting pressure variations near boundaries and high-gradient regions, indicates that BINN effectively captures the flow physics induced by full-tensor permeability.

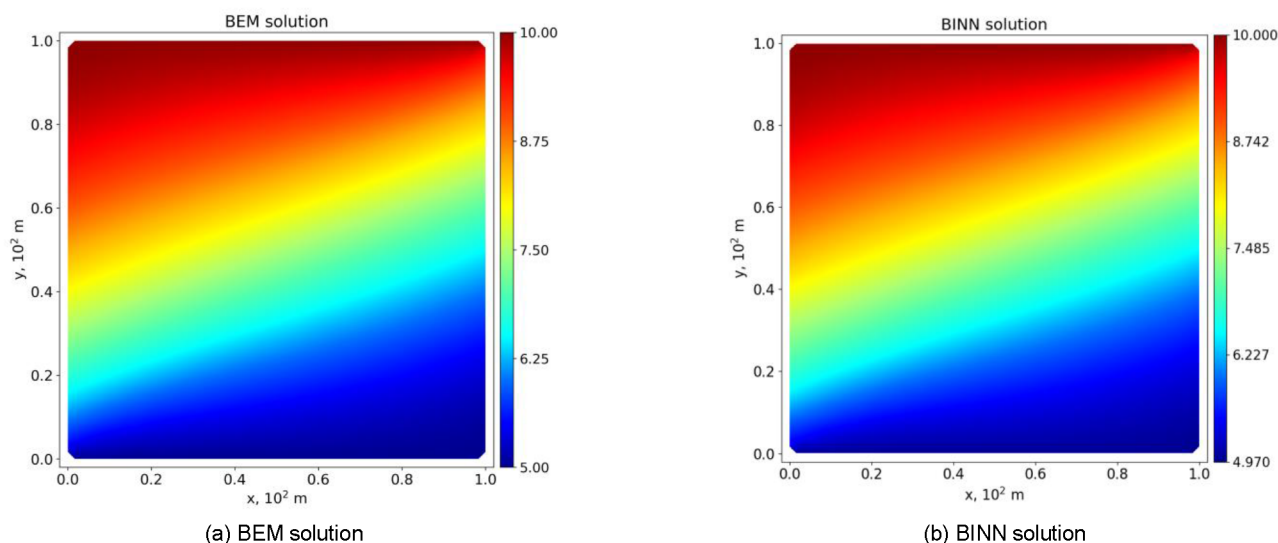


Figure 5—Comparison of pressure field distributions between BINN solution and BEM solution.

Error analysis further validates BINN's superiority. Fig. 6 shows that the absolute error magnitude across the entire computational domain is generally below 7.22×10^{-2} MPa, with an average error of 5.73×10^{-3} MPa and root mean square error (RMSE) of 9.66×10^{-3} MPa. The error distribution exhibits a uniform spatial distribution without pronounced local accumulation, underscoring the method's consistency in

accuracy. By integrating the physical constraints of boundary integral equations with the learning capability of neural networks, BINN provides an efficient and reliable technical pathway for solving seepage problems in anisotropic reservoirs.

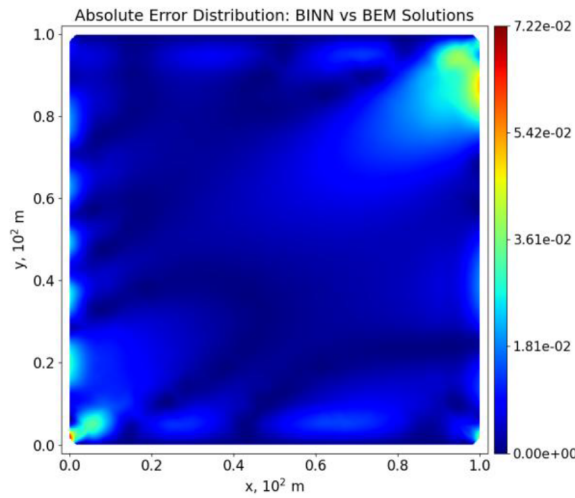


Figure 6—Absolute error distribution of BINN solution and BEM solution (unit: MPa)

Example 2

This example further introduces the source-sink phase based on Example 1, in order to investigate BINN in the simulation of production wells in reservoirs. The study domain is still a rectangular reservoir with dimensions of 100m×100m. Anisotropic permeability tensor

$\mathbf{K} = \begin{pmatrix} 500 & 0 \\ 0 & 200 \end{pmatrix} mD$ is considered, and the fluid viscosity $\mu = 1 \text{ mPa} \cdot \text{s}$. All four boundaries of the reservoir are set as Dirichlet boundaries, with a uniform pressure of $p = 10 \text{ MPa}$, that is:

$$\begin{cases} p(x, 0) = 10\text{MPa}, & 0 \leq x \leq 100 \\ p(x, 100) = 10\text{MPa}, & 0 \leq x \leq 100 \\ p(0, y) = 10\text{MPa}, & 0 \leq y \leq 100 \\ p(100, y) = 10\text{MPa}, & 0 \leq y \leq 100 \end{cases} \quad (22)$$

There is a production well (source-sink term) at the center point (50, 50), with an intensity of $Q(50,50) = -500 \times 0.0864 \text{ m}^3/\text{day}$, and $Q(\mathbf{x}) = 0 \text{ m}^3/\text{day}$ in the remaining areas (Fig.7). This setup simulates the situation where the existence of the production well causes the internal pressure field to form a pressure drop funnel around the center point.

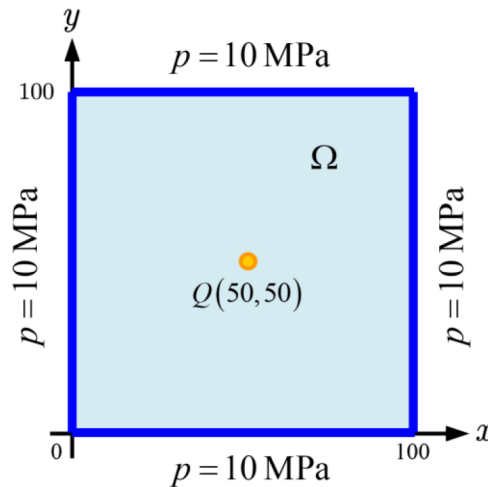


Figure 7—Computational domain geometry and boundary conditions.

The boundaries are discretized using the same constant element scheme as in Example 1, with each of the four sides is divided into 30 elements, resulting in a total of 120 boundary elements. The midpoints of the elements are used as collocation points. 400 internal points are selected for the visualization and accuracy verification of the pressure field. The neural network architecture in Example 2 follows the identical configuration as that in Example 1 as well.

The variation of the loss function in Fig. 8 demonstrates that the loss value decreases rapidly and tends to stabilize during the iteration process, indicating that the method exhibits excellent convergence in handling boundary integral equations and can effectively approximate the boundary pressure distribution through an adaptive learning mechanism.

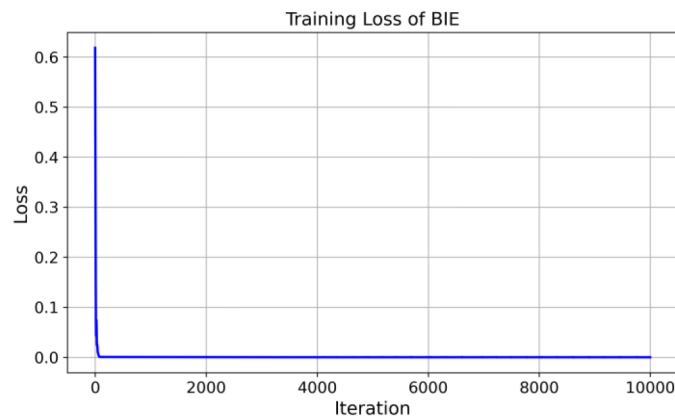


Figure 8—Loss function of BIE in Example 2.

The pressure distribution results in Fig. 9 show that both the method and the BEM (reference solution) can capture the pressure drop funnel feature around the production well. Due to the anisotropic permeability, the pressure drop expands more rapidly along the $-x$ -axis, forming an elliptical distribution. The pressure contour lines of the two methods are in high agreement, and the pressure decay law is accurately characterized in the high-gradient regions near the boundary and the well, reflecting the effective capture of the seepage physics.

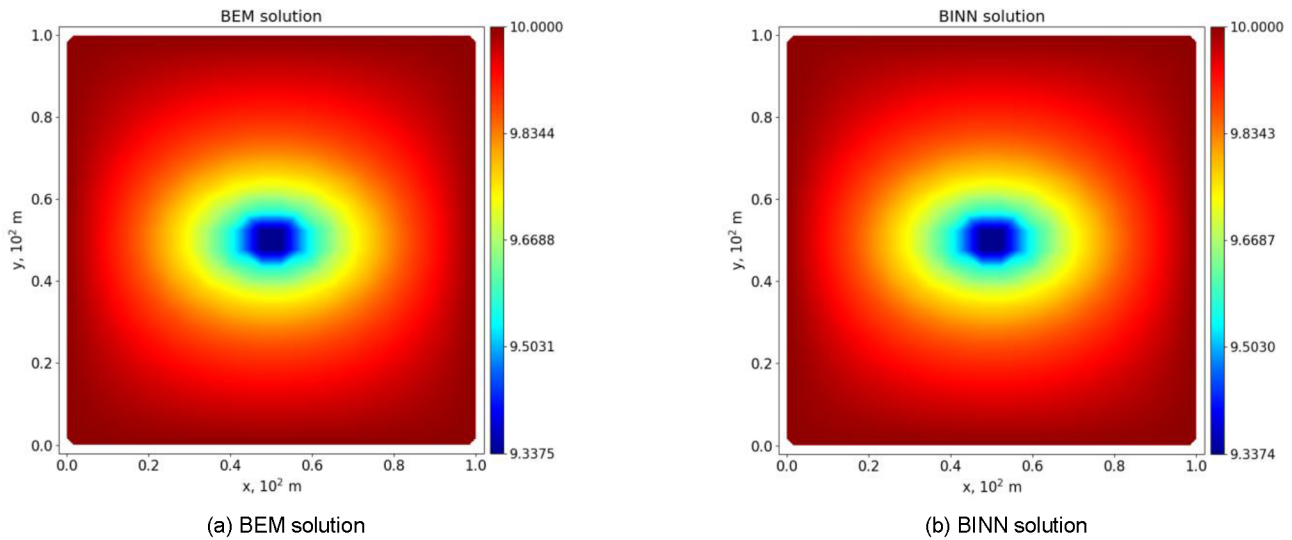


Figure 9—Comparison of pressure field distributions between BINN solution and BEM solution.

Fig. 10 presents the spatial distribution characteristics of the error between the solution of this method and the BEM reference solution. The results show that the absolute error magnitude in the entire computational domain is generally lower than 2.62×10^{-3} MPa. The average error is 3.03×10^{-4} MPa, and the RMSE is 5.17×10^{-4} MPa. Notably, even in the region with the steepest pressure gradient at the well center, the error remains far within the engineering acceptable threshold, highlighting the method's remarkable precision in critical high-gradient zones. This error analysis indicates that BINN can naturally and highly accurately handle the singular source-sink terms caused by injection-production wells in anisotropic reservoir flow.

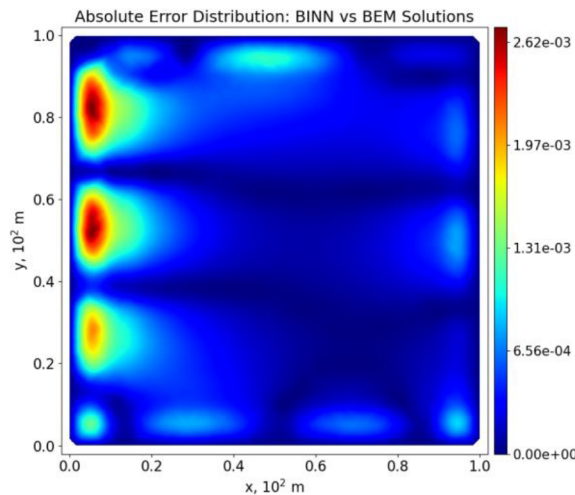


Figure 10—Absolute error distribution of BINN solution and BEM solution (unit: MPa)

Example 3

This example investigates seepage with recessed boundaries and multi-well coupling. The computational domain is a trapezoid (lower base 220m, upper base 130m, height 120m) featuring a 75m-deep, 70m-wide rectangular recess at the center of its lower base (Fig. 11), distinct from the regular rectangular domains in prior examples. A full-tensor permeability $\mathbf{K} = \begin{pmatrix} 300 & 80 \\ 80 & 100 \end{pmatrix}$ mD and $\mu = 1$ mPa $\cdot$ s are adopted, with all four boundaries set to a constant pressure of 10 MPa. Crucially, two production wells are placed at [40,50]m

and [180,70]m with intensities of $-300 \text{ m}^2/\text{s}$ and $-200 \text{ m}^2/\text{s}$, simulating asymmetric pressure fields under multi-well interference in realistic reservoirs.

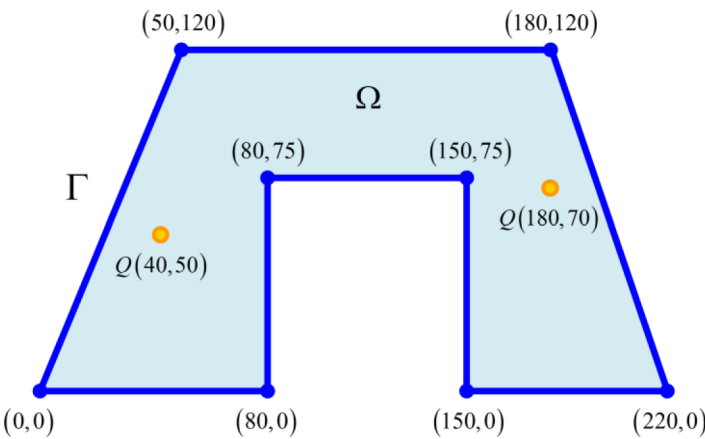


Figure 11—Computational domain geometry and boundary conditions.

Boundary discretization employs constant elements. Due to geometric irregularity, a 5m spacing generates 153 boundary elements. 1400 interior points are arranged for pressure field reconstruction and accuracy verification.

The BINN architecture is enhanced for physical complexity: hidden layers increase to 4 with 60 neurons per layer to strengthen nonlinear mapping. The optimizer, activation function, learning rate, and iterations remain consistent with previous examples to ensure comparative fairness shown in Table 2. This example focuses on validating BINN's capability to resolve pressure gradients under the coupled effects of full-tensor permeability, complex boundary geometry, and multi-well interference.

Table 2—Parameters of the neural networks and optimizers in Example 3

Model	Number of hidden layers	Number of neurons per hidden layer	Total number of parameters	Activation Function	Optimizer (Adams)
BINN	4	60	11221	Tanh	10000 iterations, learning rate 10^{-3}

Fig. 12 demonstrates the evolution of the loss function during BINN training iterations. The loss value first drops rapidly and then gradually converges and stabilizes, indicating that even with the geometric singularities caused by the recessed boundaries, BINN still demonstrates strong good convergence in solving boundary integral equations.



Figure 12—Loss function of BIE in Example 3.

Pressure distributions in Fig. 13 reveal that both the BINN solution and the BEM solution can accurately present the asymmetric pressure funnel morphology caused by the interference of two wells. It is worth noting that the elliptical pressure drop areas around each well will preferentially extend along the principal permeability direction. Moreover, in sensitive areas such as wellbores and recess vertices, the pressure field distributions of the two methods are highly consistent, which fully demonstrates that BINN can effectively capture the flow physics under the coupling of full-tensor anisotropic permeability, boundary-induced flow, and multi-well superposition effects.

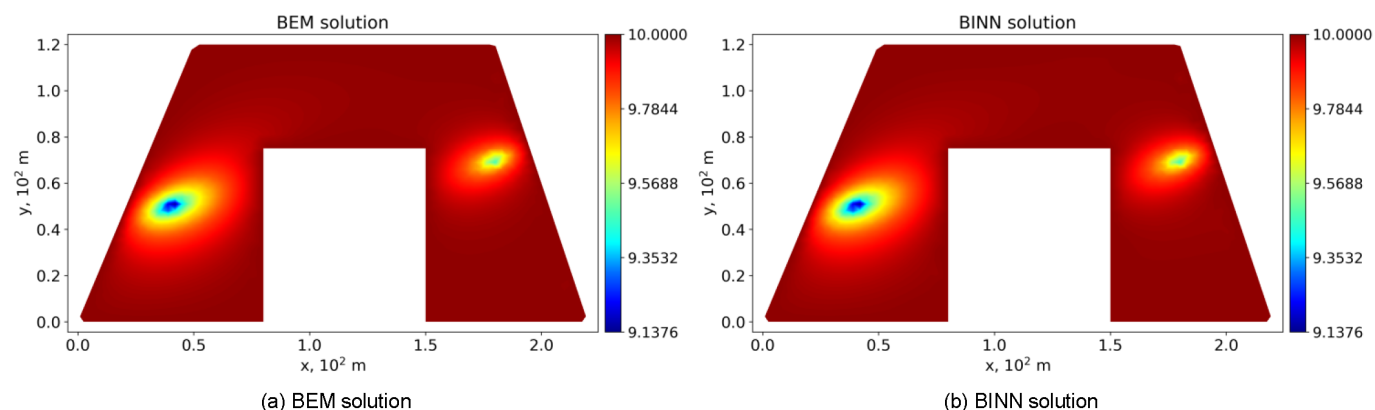


Figure 13—Comparison of pressure field distributions between BINN solution and BEM solution.

Error quantification substantiates BINN's engineering reliability. As depicted in Fig. 14, the absolute error in the entire computational domain is always controlled below 8.82×10^{-3} MPa, and the error is relatively uniform in spatial distribution without serious error accumulation. Although the geometry and physics of this example are more complex than the previous two examples, the root mean square error is still less than 10^{-3} MPa, which fully verifies that BINN has good scalability in handling complex seepage problems.

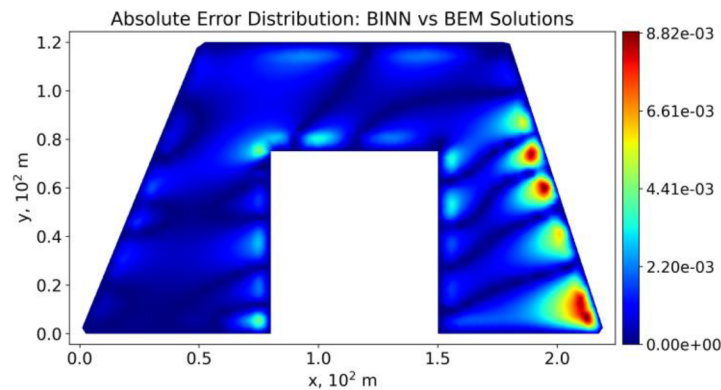


Figure 14—Absolute error distribution of BINN solution and BEM solution

Conclusion

This study presents a Boundary-Integral Neural Network (BINN) framework for simulating single-phase flow in anisotropic reservoirs, with its performance systematically validated through three representative examples. By integrating boundary integral equations (BIEs) with deep learning, BINN effectively addresses the limitations of traditional numerical methods and Physics-Informed Neural Networks (PINNs) in anisotropic reservoir simulations. It reduces problem dimensionality by focusing on boundary unknowns, enhancing computational efficiency while avoiding mesh generation complexities. The analytical enforcement of boundary conditions ensures mathematical consistency, and the Green's function kernel naturally handles singularities near wells or fractures without special treatment.

Validation results show that BINN accurately captures pressure distribution characteristics induced by full-tensor permeability tensors, with results aligning well with Boundary Element Method (BEM) solutions and errors within engineering thresholds. It precisely simulates flow with production wells, reproducing elliptical pressure drop funnels around wells under anisotropic conditions, even in high-gradient regions (RMSE=5.17×10⁻⁴ MPa). For irregular geometries (trapezoidal domain with recesses) and multi-well interference, BINN maintains stable convergence and high accuracy (RMSE<10⁻³ MPa), confirming its scalability for complex scenarios.

Future work will extend the framework to transient flow, three-dimensional flow, multiphase flow, and fractured reservoirs to explore its applicability in more complex reservoir flow scenarios.

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